A Modular, FHIR-Integrated Framework for LLM-Assisted Pre-Consultation

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Abstract

We describe a modular framework for LLM-assisted pre-consultation with deterministic questionnaires and structured TW Core FHIR write-back. Synthetic verification covered interoperability, text-channel workflow, model replacement, and local load; clinician confirmation was simulated. Across the selected E1 cohorts, 1,347 written resources, including 350 QuestionnaireResponses, had 0 errors and 0 validator exceptions, meeting the zero-error target. End-to-end completion was 100.0% (150/150) across 150 sessions. A commercial model replacement met eventual-completion and FHIR criteria; local Gemma retained only 6 chief-complaint facts and completed without handoff in 93.3% (28/30), failing the full replacement criterion. Mock load at 500 concurrent sessions had

74.0% session failure despite 4.71% request error. Small real-model load completed 7 of 7 sessions. Exchange-hub testing (E5) remains blocked by access. Physician scenario review and independent audits are pending. These findings establish bounded software verification, not clinical effectiveness or deployment readiness.

Keywords: FHIR, Pre-consultation, Conversational AI, Clinical Documentation

1 Introduction

Low-acuity visits place a growing burden on unscheduled care worldwide.Pines et al. (2011) In Taiwan, despite low-cost access under National Health Insurance, its emergency departments remain chronically crowded, and about one-third of visits are non-emergent.Huang et al. (2019) This crowding prolongs waits, worsens outcomes, and drives clinician burnout.Morley et al. (2018); Ghaseminejad et al. (2025)

Large language models (LLMs) now support numerous clinical tasks, including ambient scribing, EHR documentation, and clinical decision support.Nair et al. (2026) Agentic systems extend these capabilities through planning, tool use, and multi-step action, enabling closer integration with health-system workflows and conversational diagnosis.Tu et al. (2025) However, although various benchmarks now evaluate LLMs on realistic tasks and interactive encounters, most agentic systems remain confined to simulated settings.Liu et al. (2026)

Here, we present XXX, an LLM-based agent for adults presenting in Taiwanese urgent care centers. XXX conducts structured history-taking by text or voice, drafts a clinician-ready note, and supports decision-making with a ranked differential and not-to-miss conditions, alongside suggested examinations and imaging. To limit hallucination, the LLM only extracts clinical facts linked to the patient's own words, while safety screening, question selection, and diagnostic ranking follow auditable rules. XXX connects to EHRs via SMART on FHIR and was evaluated on benchmark.

2 Methods

2.1 Scope and implemented architecture

We verified a reference implementation for structured pre-consultation using synthetic patients. The outcomes concern data interchange, interview-to-write-back completion, model replacement, and local concurrency. Diagnostic and triage accuracy, clinical note quality, and comparisons with clinicians were outside scope.

The implementation combines Vue patient and clinician interfaces, a FastAPI backend, local retrieval, and a HAPI FHIR sandbox (Fig. 1) (HAPI FHIR Project 2026). A patient invitation is exchanged for a scoped session cookie; the clinician launcher binds patient and encounter identifiers. The separate SMART integration surface follows a different authorization flow (HL7 International 2024). These experiments exercise text-channel HTTP APIs, not browser/OAuth launch or voice input.

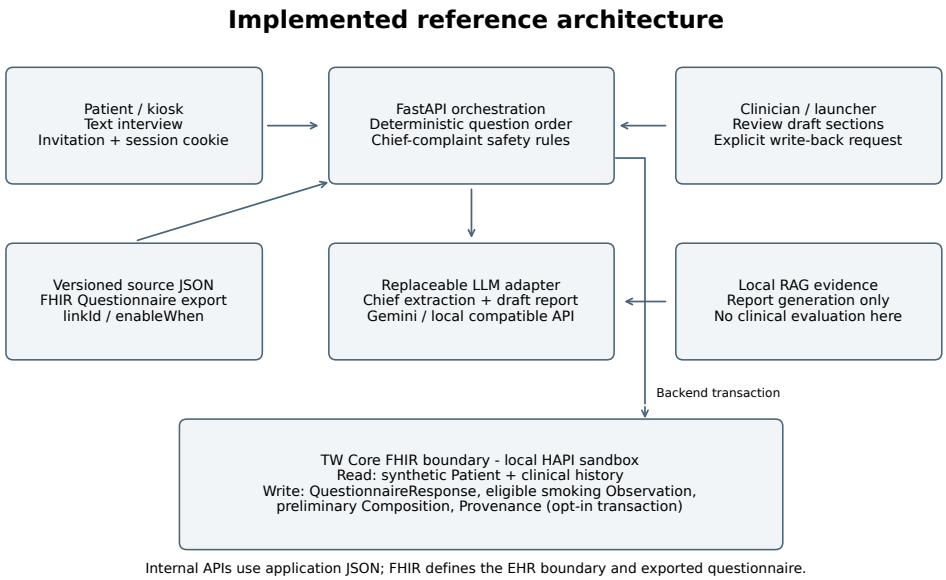


Fig. 1 Implemented modules and data boundaries. Source JSON controls the questionnaire engine; a corresponding FHIR Questionnaire is exported. Internal APIs use application JSON, while FHIR defines the EHR boundary.

Isolated experiment databases and an explicit loopback development authorization bypass separate the runs from production services.

Question selection is deterministic: the engine asks the next enabled question and skips EHR-prefilled fields. Rules and evidence-checked model extraction screen the chief complaint; failed semantic assessment triggers manual handoff. At routine completion, retrieval supplies evidence for model-generated report sections and decision-support suggestions. These suggestions are neither validated diagnoses nor published orders.

2.2 Questionnaire and write-back contracts

The encoder maps source fields to stable `linkId` values, options to coded answers, and supported conditions to `enableWhen`. Questionnaire and QuestionnaireResponse separate form definitions from answers (HL7 International 2019, 2022). The canonical version includes a source digest and is pinned in the interview trace when a route is selected. Write-back rejects a changed canonical version; older records without a pin use current resolution with an explicit legacy tag. Application extensions carry source and presentation semantics, so generic FHIR consumers cannot assume complete runtime equivalence.

Table 1 Write-back resources and assertion boundaries.

Resource	Content and boundary
QuestionnaireResponse	Ordered answers, source markers, and the pinned canonical reference.
Observation	Smoking status only for an explicit mapped patient selection; EHR text is not promoted to a new patient assertion.
Composition	Preliminary narrative, confirmed sections, and links to structured answers.
Provenance	Patient informant, AI assembly, and simulated clinician-review roles; no actual clinical approval.

Structured write-back is opt-in; Composition-only write-back remains the default. After simulated clinician confirmation, an authorized request creates the transaction in Table 1. Stable references and conditional creation support retry idempotency. Patient, encounter, institution, and issuer checks precede persistence. No differential-diagnosis Conditions, ClinicalImpressions, ServiceRequests, or safety Flags are created.

An absent-EHR placeholder is omitted from the response. Free text outside the option grammar is represented by the existing coded "other" option plus its nested string, retaining both the original text and answer-source marker. This avoids the validator's incompatible open-choice string/coded-condition exception without inventing a clinical selection. The original minimal reproduction is retained. When an answered runtime branch would lose its answer in FHIR, write-back fails closed with a non-retryable unprocessable-content response; unanswered branches do not create answers or block writing.

2.3 Synthetic inputs and model disclosure

Synthea generated the adult source cohort (Walonoski et al. 2018; The MITRE Corporation 2026). Conversion assigns fictional Taiwanese identifiers and addresses and applies TW Core profiles (Ministry of Health and Welfare, Taiwan 2025). Seeds, mappings, dropped elements, and content hashes are recorded. Taiwan-styled presentation does not establish local disease representativeness. Input validation is separate from system-output validation. The 30 author-drafted scenarios cover chest pain, headache, and abdominal pain, including expected safety stops and known rule gaps. The planned benchmark was unavailable. Separate source-level Synthea modules provide scenario-consistent backgrounds; generated patient records are not manually rewritten to force agreement.

Mechanical audits check conditions, medications, allergies, and smoking history. Human approval requires two independent physician decisions against current script hashes, including patient bindings and expected paths. At generation, 0 scripts have those approvals. Model and sampling settings are disclosed by role in Supplementary Tables 12 and 16. Commercial models use remote APIs; local Gemma and Qwen use Ollama (Ollama 2026; Gemma Team et al. 2025; Qwen Team 2025). Requested temperature is distinguished from the transmitted value: the Gemini adapter omits

temperature for the configured newer Gemini models. Provider-effective sampling and unrecorded historical thinking settings are not inferred from defaults. Implementation, scenario drafting, orchestration, reference checking, and manuscript revision used AI assistance; the declarations specify the remaining author review.

2.4 Verification procedures

E1 uses the pinned HL7 validator with TW Core and generated questionnaire and extension definitions (HL7 International 2026; Ministry of Health and Welfare, Taiwan 2025). Structural/profile checks use `-tx n/a`; they do not establish complete terminology resolution. The prespecified target is no errors on written resources. Missing outcomes and validator exceptions are failed verification, reported separately from profile errors. Round-trip comparison ignores specified server metadata and assigned identifiers but preserves substantive differences. The independent E1 cohort uses the same seeded patient selection and fixed short complaints as the earlier run. A supplemental E1 validates the new E2 outputs without double counting.

The pinned TW Core Provenance binding references participant-type CodeSystem version 4.0.1, unavailable in the local terminology cache. Explicitly substituting a different cached version produced binding errors in a diagnostic fixture; we therefore retained the correct system URI and reported the warnings. Composition narrative warnings are also retained (Supplementary Table 10).

E2 runs the patient-facing API through invitation, dialogue, report generation, simulated clinician confirmation, transaction, and read-back. A distinct local Qwen model plays the patient using only scenario facts. Invalid constrained answers receive an explicit unknown fallback, never an expected-answer substitution. Completion requires successful write-back and an explicit `completed` or `summary_ready` workflow. This includes successful chief-complaint safety stops and recovery after a timeout, but excludes manual handoff. We separately report completion without observed timeout. Timing retains all API write-backs. Protocol adherence excludes expected red-flag chief stops from comparable paths; unexpected stops remain failures in the denominator. Traceability uses every persisted chief-complaint fact, including any missing evidence, and checks its span against patient utterances. Functional tests compare rule identifiers; they are not sensitivity or specificity measurements.

E3 changes only module configuration and repeats workflow and FHIR checks. A complete replacement requires completion without handoff and successful FHIR verification; successful model invocation alone is insufficient. One-time adapter implementation is measured separately against base commit `1b0eb1c`. Safe session/request identifiers link parser counts, post-validation counts, schema failures, and bounded-operation timeouts. Candidate-check failures are not unique rejected facts. Local system and simulated-patient models share GPU resources. E4 replays captured turns with either fixed-latency mock responses or real models, using backend v0.18.0 after this fix round and one worker. The prespecified stop criterion remains request error rate above five percent; session failure is a separate endpoint. The asynchronous HTTP harness replaces the planned k6/Locust tool. E5 remains blocked pending exchange-hub access.

Table 2 E1 written-resource validation. Input checks are excluded; exceptions would count as failed verification.

Cohort	Resource	n	With errors	Errors	Warnings
Templated cohort	Composition	200	0	0	200
Templated cohort	Observation	197	0	0	0
Templated cohort	Provenance	200	0	0	800
Templated cohort	QuestionnaireResponse	200	0	0	0
Scenario cohort	Composition	150	0	0	150
Scenario cohort	Observation	100	0	0	0
Scenario cohort	Provenance	150	0	0	600
Scenario cohort	QuestionnaireResponse	150	0	0	0

Table 3 E1 workflow and semantic round trip. Read uses all sessions; write uses attempted writes; read-back uses returned references; round trip uses all captured submissions, including those without read-back.

Cohort	Sessions	Read	Write	Read-back	Round-trip
Templated cohort	200	100.0% (200/200)	100.0% (200/200)	100.0% (797/797)	100.0% (797/797)
Scenario cohort	150	100.0% (150/150)	100.0% (150/150)	100.0% (550/550)	100.0% (550/550)

Results are descriptive; repetitions are not independent clinical observations. Tables show actual denominators and median/interquartile summaries. Cost covers bounded interactive request calls only: background summaries and late calls can be absent, and every usage record declares incomplete coverage. A successful recorded call can coexist with an outer operation timeout. Source hashes and explicit run selection generate all result macros, tables, and figures.

3 Results

3.1 Interoperability

Independent validation of the converted input cohort covers 1,000 bundles and 430,017 resources, with 0 errors and 1,058,802 warnings. This combines unchanged outcomes with hash-checked revalidation of modified bundles and is separate from system-output conformance.

E1 includes 200 independent input-cohort sessions and 150 E2 sessions. The 1,347 written resources have 0 errors and 0 validator exceptions. All 350 QuestionnaireResponses have explicit outcomes, with 0 errors and 0 unverified responses. The zero-error target is met in both cohorts (Tables 2 and 3). The retained 1,750 warnings comprise 1,400 Provenance terminology warnings and 350 Composition narrative warnings. The previous QR verification gap is resolved by faithful nested-text encoding.

Table 4 E2 software workflow outcomes. Percentages show actual numerators and denominators; path comparison excludes expected chief stops.

Cohort	n	Completion	Path exact	Traceable facts	Turns median [Q1,Q3]	Median seconds [Q1,Q3]
Chest pain	50	100.0% (50/50)	100.0% (35/35)	100.0% (144/144)	15.5 [1.0, 17.0]	35.9 [12.9, 40.3]
Headache	50	100.0% (50/50)	100.0% (30/30)	100.0% (152/152)	14.5 [1.0, 15.0]	33.9 [9.1, 38.3]
Abdominal pain	50	100.0% (50/50)	100.0% (35/35)	100.0% (158/158)	13.0 [1.0, 15.0]	32.7 [13.4, 36.6]
Overall	150	100.0% (150/150)	100.0% (100/100)	100.0% (454/454)	15.0 [1.0, 15.0]	34.6 [12.1, 38.9]

Table 5 E2 functional-test pass rates and bounded observed usage. Costs exclude local hardware and unrecorded calls.

Overall provisional measurement	Observed value
Red-flag functional-test pass rate: exact rules	83.3% (50/60)
Red-flag functional-test pass rate: acceptable rules	83.3% (50/60)
Other-script functional-test pass rate: no unexpected rules	100.0% (90/90)
Sessions with documented rule-coverage gaps	10
Sessions with recorded model usage	150
System input tokens/session, median	17,014
System output tokens/session, median	1,776
Simulated-patient tokens/session, median	17,522
Observed API cost/session, median USD	0.0177
Sessions contributing cost estimates	110
Invalid simulated-patient turns	48

3.2 Workflow and model replacement

E2 contains 150 sessions, with 5 repetitions per script. Completion is 100.0% (150/150), path agreement is 100.0% (100/100), and fact traceability is 100.0% (454/454) (Table 4). All 50 chief stops completed write-back. The 48 invalid simulated-patient turns used explicit unknown fallbacks. There were 1 timeout-affected sessions; 1 retained a record and 1 eventually completed after retry. Completion without observed timeout was 149 of 150. Table 5 reports functional checks and partial usage; independent audit fields remain blank.

The 3 E3 configurations used identical executable fingerprints throughout. One-time LLM adapter work added 682 and removed 60 lines relative to the base commit, restricted to the adapter-related files; subsequent configuration swaps required no code changes. The commercial alternate met eventual-completion and FHIR criteria, although 2 sessions recovered after timeouts. The local Gemma replacement did not: completion without handoff was 93.3% (28/30), despite successful FHIR checks (Table 6). Its unexpected stops remain path failures; only expected stops are excluded from that denominator.

Persisted facts were 91 for the default, 114 for the commercial alternate, and 6 for Gemma. Gemma's 22 extraction calls produced 108 parsed symptom candidates but retained only 6; findings were absent before and after validation. No local

Table 6 E3 model replacement. Code changes exclude one-time adapter work. Completion excludes handoff. Costs are partial API-charge estimates; local zero excludes hardware and omitted calls.

Configuration	Code lines	Completion	Path exact	Facts	Errors	Median seconds	USD/session*
Default Gemini	0	100.0% (30/30)	100.0% (20/20)	100.0% (91/91)	0	40.0	0.0192 (n=22)
Alternate Gemini	0	100.0% (30/30)	100.0% (20/20)	100.0% (114/114)	0	34.5	0.0096 (n=22)
Local Gemma 3	0	93.3% (28/30)	90.9% (20/22)	100.0% (6/6)	0	205.7	0.0000 (n=20)

schema/parser failures were observed (Table 7). AP-07 and HA-08 did not stop at chief screening; HA-08 retained only a headache fact. AP-03 and AP-04 entered manual handoff after chief-assessment timeouts; late worker completion did not reverse these outcomes. Gemma's 12 timeout-affected sessions all retained records, but only 10 completed without handoff. All 311 captured usage calls reported `ok`; this bounded lower-level status did not describe the timed-out outer operations. The earlier excluded run lacks pre-validation diagnostics, preventing precise retrospective attribution of its low extraction yield.

3.3 Local load

Mock load at concurrency 100 had p95 per-turn latency of 121.3 seconds. At 500, 130 of 500 sessions completed; request error was 4.71% and session failure 74.0%. The request-error stopping threshold was not crossed despite high session failure. The lower p95 at the highest level is subject to survivorship: sessions failing early contribute no later-turn latency. It does not establish faster successful workflows (Fig. 2; Supplementary Table 9). Real mode completed 7 of 7 sessions at concurrency 1, 2, 4. Both modes replay the selected E2 source but use different load levels and patient pools; these are local descriptive measurements. National peak demand remains `TODO_PEAK_DEMAND`. E5 has no measured result.

4 Discussion

Verification distinguishes data fidelity from successful transport. Nested "other" encoding preserved unrecognized EHR text while resolving QR validation exceptions. Version pinning and answered-branch guards make incompatible changes explicit. These contracts are narrower than a universal FHIR-native internal bus: source JSON still controls interviews and extensions carry application semantics.

Model configurability likewise does not guarantee equivalent behavior. Gemma's low fact yield arose with nonempty symptom candidates that disappeared during local code/evidence validation, absent findings, and separate operation timeouts. The diagnostics do not retain raw candidate JSON, so they cannot allocate each removed candidate to incorrect codes, unsupported evidence, or deduplication. A traceability

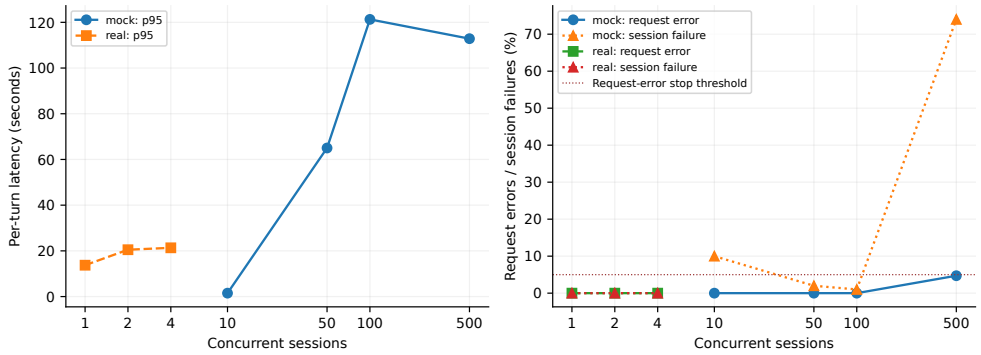


Fig. 2 Local synthetic load on the fixed backend. The dotted line is the request-error stop threshold, not a session-failure threshold. Per-turn latency includes observed HTTP attempts; early failures truncate later measurements.

percentage over a small surviving set cannot establish coverage. The failed completion criterion and missed scripted stops preclude a claim of complete local-model interchangeability under the tested settings.

4.1 Limitations and clinical review needs

This is software verification, not evidence of clinical safety, diagnostic performance, utility, or deployment readiness. Scenarios await physician review; synthetic demographics are not representative, the patient simulator may hallucinate, and evidence spans do not validate clinical interpretation. Only chief-complaint screening runs in the default engine. Later questionnaire selections such as diaphoresis, syncope, fever with neck stiffness, unclear speech, or abdominal rigidity/rebound pain do not trigger renewed screening. Rules omit suspected pulmonary embolism (CP-09), abdominal aortic aneurysm without syncope (AP-10), and sustained exertional pressure without diaphoresis or syncope. The structured ectopic-pregnancy combination depends on prefilled pregnancy risk, although a separate raw rule can match explicit pregnancy and bleeding. The raw negation window treats single negation characters broadly: a phrase describing limb weakness followed by unclear speech can suppress the neurologic match. These code-level gaps require qualified clinical review; no rules were changed.

Cost telemetry is incomplete. Local models share GPU capacity, commercial aliases may change, and recorded versions cannot make remote inference deterministic. The sandbox does not establish production authorization, tenant isolation, browser SMART integration, or national capacity. Independent transcript and fact audits, terminology review, institutional governance, and exchange-hub access remain necessary before further acceptance.

5 Conclusion

The fixed implementation meets the written-resource verification target and supports inspectable questionnaire, provenance, and write-back contracts. Model replacement and load results expose operational limits that successful FHIR transfer alone would conceal. Clinical evaluation remains separate.

Supplementary information. The repository preserves immutable runs, source hashes, scripts, reviewer packets, and analysis code. The independent E1 cohort rotates fixed brief complaints across seeded patients. The harness composes option answers as the kiosk does, records invalid attempts, and uses exact unknown-option matching; an earlier substring fallback was excluded because it could select a symptom-containing option. Run-created resources are captured and removed before patient reuse. Shared patients are serialized within a run, and experiments run sequentially after baseline checks. No current chest-pain scenario has human approval, so a clinician-approved walkthrough remains unavailable.

Table 7 separates chief stops, extraction yield, and timeouts; Table 8 gives each local-model session. Timeout logs and bounded-operation events can describe the same incident and are not added. Parser/shape failures are local observations, not proof of remote schema enforcement.

Acknowledgements. Author confirmation of acknowledgements and institutional wording is pending.

Declarations

Funding, competing interests, and contributions. Authors must supply and verify these statements and final approval.

Ethics and consent. Only synthetic data were used. Authors must confirm the applicable institutional determination; no ethics approval is asserted.

Data and code availability. Scripts, derived summaries, and generators are in the project repository. Large raw artifacts are regenerable and excluded from source control. An archival identifier and access statement require author confirmation.

AI assistance. Claude Code and Codex assisted code and scenario drafting, experiment orchestration, source checking, and manuscript drafting and revision. Recorded model identifiers, dates, and unavailable settings are disclosed in the repository. Authors must verify citations, review technical and clinical content, and approve the final text; scenario approvals and independent audits remain pending.

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Table 7 E3 extraction and operation diagnostics. Expected chief stops alone are excluded from path comparison; unexpected stops remain failures.

Observed measurement	Default Gemini	Alternate Gemini	Local Gemma 3
Sessions	30	30	30
Persisted facts	91	114	6
Chief stops (all)	10	10	10
Chief stops with eventual completion	10	10	8
Unexpected chief stops	0	0	2
Eventual completion (no handoff)	100.0% (30/30)	100.0% (30/30)	93.3% (28/30)
Manual handoffs	0	0	2
Expected chief stops excluded from path test	10	10	8
Red-flag functional-test pass rate	83.3% (10/12)	83.3% (10/12)	66.7% (8/12)
Sessions with observed timeout	0	2	12
Timeout sessions: stored record	0	2	12
Timeout sessions: API write-back	0	2	12
Timeout sessions: eventual completion	0	2	10
Completion without observed timeout	30	28	18
Bounded-operation timeout events	0	2	12
Timeout log events (not additive)	0	2	10
Observed chief extraction calls	22	22	22
Local parser/shape failures	0	0	0
Symptoms: parsed to post-validation	$22 \rightarrow 22$	$28 \rightarrow 22$	$108 \rightarrow 6$
Findings: parsed to post-validation	$52 \rightarrow 51$	$53 \rightarrow 53$	$0 \rightarrow 0$
Assessments: parsed to post-validation	$22 \rightarrow 22$	$28 \rightarrow 22$	$22 \rightarrow 0$
Questionnaire answers: parsed to post-validation	$45 \rightarrow 52$	$67 \rightarrow 70$	$0 \rightarrow 20$

Table 8 Gemma session diagnostics. Symptoms show parsed to retained candidates; "–" means no extractor call. Handoff* includes worker events occurring after session completion; these do not undo handoff. Empty usage is not zero consumption.

Script	Workflow	Facts	Symptoms parsed → retained	Findings parsed → retained	Schema failures	Timeout (events)	Usage status/calls
AP-01	Completed	0	3 → 0	0 → 0	0	RAG (1)	ok/16
AP-02	Completed	0	3 → 0	0 → 0	0	RAG (1)	ok/15
AP-03	Handoff*	0	3 → 0	0 → 0	0	Chief (1)	Empty/0
AP-04	Handoff*	0	3 → 0	0 → 0	0	Chief (1)	Empty/0
AP-05	Completed	0	3 → 0	0 → 0	0	None	ok/16
AP-06	Completed	0	3 → 0	0 → 0	0	None	ok/16
AP-07	Completed	0	3 → 0	0 → 0	0	RAG (1)	ok/15
AP-08	Chief stop	0	– → –	– → –	0	None	Empty/0
AP-09	Chief stop	0	– → –	– → –	0	None	Empty/0
AP-10	Completed	0	3 → 0	0 → 0	0	RAG (1)	ok/15
CP-01	Completed	0	11 → 0	0 → 0	0	RAG (1)	ok/15
CP-02	Completed	0	11 → 0	0 → 0	0	None	ok/16
CP-03	Completed	0	11 → 0	0 → 0	0	RAG (1)	ok/16
CP-04	Completed	0	11 → 0	0 → 0	0	RAG (1)	ok/15
CP-05	Completed	0	11 → 0	0 → 0	0	None	ok/16
CP-06	Completed	0	11 → 0	0 → 0	0	None	ok/16
CP-07	Chief stop	0	– → –	– → –	0	None	Empty/0
CP-08	Chief stop	0	– → –	– → –	0	None	Empty/0
CP-09	Completed	0	11 → 0	0 → 0	0	RAG (1)	ok/14
CP-10	Chief stop	0	– → –	– → –	0	None	Empty/0
HA-01	Completed	1	1 → 1	0 → 0	0	None	ok/16
HA-02	Completed	1	1 → 1	0 → 0	0	None	ok/16
HA-03	Completed	1	1 → 1	0 → 0	0	None	ok/16
HA-04	Completed	1	1 → 1	0 → 0	0	RAG (1)	ok/15
HA-05	Completed	1	1 → 1	0 → 0	0	RAG (1)	ok/15
HA-06	Completed	0	1 → 0	0 → 0	0	None	ok/16
HA-07	Chief stop	0	– → –	– → –	0	None	Empty/0
HA-08	Completed	1	1 → 1	0 → 0	0	None	ok/16
HA-09	Chief stop	0	– → –	– → –	0	None	Empty/0
HA-10	Chief stop	0	– → –	– → –	0	None	Empty/0

Table 9 Every measured load level. Latency is per observed turn; completion and request errors have different denominators.

Mode	Concurrency	Completed	Request error	Session failure	Turn p95 (s)
Mock	10	9/10	0.00%	10.00%	1.5
Mock	50	49/50	0.00%	2.00%	65.0
Mock	100	99/100	0.00%	1.00%	121.3
Mock	500	130/500	4.71%	74.00%	112.8
Real	1	1/1	0.00%	0.00%	13.7
Real	2	2/2	0.00%	0.00%	20.5
Real	4	4/4	0.00%	0.00%	21.3

Table 10 Retained E1 warnings across both written-resource cohorts.

Count	Warning
1,400	A definition for CodeSystem 'http://terminology.hl7.org/CodeSystem/provenance-participant-type' version '4.0.1' could not be found, so the code cannot be validated. Valid versions: 0.1.0, 1.1.0 or 2.0
350	Constraint failed: dom-6: 'A resource should have narrative for robust management' (defined in http://hl7.org/fhir/StructureDefinition/DomainResource) (Best Practice Recommendation)

Table 11 E2 restricted to currently physician-approved script hashes. No approved scripts are available; absent results are not zero-valued successes.

Cohort	n	Completion	Path exact	Traceable facts	Turns median [Q1,Q3]	Median seconds [Q1,Q3]
Chest pain	0	–	–	–	–	–
Headache	0	–	–	–	–	–
Abdominal pain	0	–	–	–	–	–
Overall	0	–	–	–	–	–

Table 12 Models by executed configuration and role. Version strings are pinned metadata; dates are UTC. A configured role need not run in every session.

Configuration / role	Provider / access	Model ID / recorded version	Run dates (UTC)
Default Gemini: Chief extraction	Gemini, remote API	gemini-3.6-flash 3.6-flash-07-2026	2026-10-13
Default Gemini: Summary extraction	Gemini, remote API	gemini-3.5-flash-lite 3.5-flash-lite-07-2026	2026-10-13
Default Gemini: Summary reasoning	Gemini, remote API	gemini-3.6-flash 3.6-flash-07-2026	2026-10-13
Default Gemini: RAG translation	Gemini, remote API	gemini-3.5-flash-lite 3.5-flash-lite-07-2026	2026-10-13
Default Gemini: Simulated patient	Ollama, local API	qwen2.5:7b-instruct 845dbda0ea48ed749caafd9e 6037047aa19acfcfd82e704d 7ca97d631a0b697e	2026-10-13
Alternate Gemini: Chief extraction	Gemini, remote API	gemini-2.5-flash 001	2026-10-13
Alternate Gemini: Summary extraction	Gemini, remote API	gemini-2.5-flash 001	2026-10-13
Alternate Gemini: Summary reasoning	Gemini, remote API	gemini-2.5-flash 001	2026-10-13
Alternate Gemini: RAG translation	Gemini, remote API	gemini-3.5-flash-lite 3.5-flash-lite-07-2026	2026-10-13
Alternate Gemini: Simulated patient	Ollama, local API	qwen2.5:7b-instruct 845dbda0ea48ed749caafd9e 6037047aa19acfcfd82e704d 7ca97d631a0b697e	2026-10-13
Local Gemma: Chief extraction	Ollama, local API	gemma3:12b b92b218582c521ba287f38f1 d6057dcd3c6cbff036f13cac 97b5d861f5e58ec3	2026-10-13
Local Gemma: Summary extraction	Ollama, local API	gemma3:12b b92b218582c521ba287f38f1 d6057dcd3c6cbff036f13cac 97b5d861f5e58ec3	2026-10-13
Local Gemma: Summary reasoning	Ollama, local API	gemma3:12b b92b218582c521ba287f38f1 d6057dcd3c6cbff036f13cac 97b5d861f5e58ec3	2026-10-13
Local Gemma: RAG translation	Ollama, local API	gemma3:12b b92b218582c521ba287f38f1 d6057dcd3c6cbff036f13cac 97b5d861f5e58ec3	2026-10-13
Local Gemma: Simulated patient	Ollama, local API	qwen2.5:7b-instruct 845dbda0ea48ed749caafd9e 6037047aa19acfcfd82e704d 7ca97d631a0b697e	2026-10-13

Table 13 Default Gemini sampling by role. Requested values are distinguished from values sent by the adapter; provider-effective defaults were not observed.

Role	Temperature: requested / sent	Seed	Token budget / thinking
Chief extraction	0 / omitted	not set	4096. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
Summary extraction	0.2 / omitted	not set	chief complaint: 240; present illness: 500; past history: 400; drug history: 400; drug allergy history: 300; personal history: 450; family history: 400. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
Summary reasoning	0.2 / omitted	not set	differential diagnoses: 900; must not miss: 1200; physical examination: 900; laboratory: 1200; imaging: 1200. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
RAG translation	0 / omitted	not set	700. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
Simulated patient	0.0 / 0.0	20261013	400. Sent unchanged.

Table 14 Alternate Gemini sampling by role. Unrecorded historical thinking settings and final request limits are explicitly unavailable.

Role	Temperature: requested / sent	Seed	Token budget / thinking
Chief extraction	0 / 0	not set	4096. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
Summary extraction	0.2 / 0.2	not set	chief complaint: 240; present illness: 500; past history: 400; drug history: 400; drug allergy history: 300; personal history: 450; family history: 400. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
Summary reasoning	0.2 / 0.2	not set	differential diagnoses: 900; must not miss: 1200; physical examination: 900; laboratory: 1200; imaging: 1200. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
RAG translation	0 / omitted	not set	700. Visible-answer budget; final request limit not recorded (thinking headroom may apply). Thinking: not recorded.
Simulated patient	0.0 / 0.0	20261013	400. Sent unchanged.

Table 15 Local Gemma system and Qwen simulated-patient sampling by role. Local API access does not imply zero hardware cost.

Role	Temperature: requested / sent	Seed	Token budget / thinking
Chief extraction	0 / 0	not set	4096. Sent unchanged.
Summary extraction	0.2 / 0.2	not set	chief complaint: 240; present illness: 500; past history: 400; drug history: 400; drug allergy history: 300; personal history: 450; family history: 400. Sent unchanged.
Summary reasoning	0.2 / 0.2	not set	differential diagnoses: 900; must not miss: 1200; physical examination: 900; laboratory: 1200; imaging: 1200. Sent unchanged.
RAG translation	0 / 0	not set	700. Sent unchanged.
Simulated patient	0.0 / 0.0	20261013	400. Sent unchanged.

Table 16 Prompt source files and planned human QA by role. Source hashes are in the disclosure artifact. Automated checks and simulated confirmation do not replace pending human review.

Role	Prompt files (hashes in disclosure JSON)	Human QA status
Chief extraction	backend/amie/chief_complaint.py backend/amie/chief_complaint_schema.py	Pending: clinical scenario review and 10% fact audit; confirmation simulated.
Summary extraction	backend/app/prompts/questionnaire_summary.py backend/app/prompts/common.py	Pending: clinical scenario review and 10% fact audit; confirmation simulated.
Summary reasoning	backend/app/prompts/questionnaire_summary.py backend/app/prompts/common.py	Pending: clinical scenario review and 10% fact audit; confirmation simulated.
RAG translation	backend/knowledge/translation.py	Pending: clinical scenario review and 10% fact audit; confirmation simulated.
Simulated patient	experiments/prompts/patient_agent_v1.md	Pending: clinical scenario review and 15% transcript audit.

Table 17 Principal pinned software versions. Full digests accompany manifests.

Component	Pinned version
Synthea	v4.0.0
FHIR	4.0.1
HAPI image	hapiproject/hapi:v8.8.0-1
TW Core IG	1.0.0
HL7 validator	6.10.4
Python	3.12.15
Node	v22.23.2
Ollama	0.32.6
default extraction	gemini-3.6-flash
default summary extraction	gemini-3.5-flash-lite
default summary reasoning	gemini-3.6-flash
swap a	gemini-2.5-flash
swap b	gemma3:12b
patient simulator model	qwen2.5:7b-instruct